

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION

WINTER SEMESTER, 2023-2024

DURATION: 1 HOUR 30 MINUTES

FULL MARKS: 75

CSE 4307: Database Management Systems

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 4 (four) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Suppose you want to build an e-commerce site or create a market place similar to Amazon. Consider each of the following points as disadvantages of keeping data in a file-processing system: 15
(CO1)
(PO1)
- Data redundancy and inconsistency,
 - Difficulty in accessing data,
 - Data isolation,
 - Integrity problem,
 - Atomicity problem,
 - Concurrent access anomalies, and
 - Security problem.
- Discuss how you can solve the problems while storing the single person data and metadata about that person (such as person's basic information, payment information, order information, information about person's customization in product choice) while ensuring protection of their personal information from third party.
- b) Differentiate between a table and a view. Can you insert data through a view? Justify your position. 10
2. a) You have to design a database for an online ride sharing system. The database must be able to keep track of the vehicle and passenger to know its location. The database will keep track of items such as drivers and passengers basic information, review score, promo code validity, payment method and information, ride history, etc. At first, list down all entities considering their functionalities. Then draw an appropriate E-R model of a new online ride sharing system, mentioning the mapping cardinalities, its minimum and maximum values, and strong and weak entity set (if any). 10
(CO2)
(PO3)
- b) Describe complexities due to the presence of ternary relationship in the E-R diagram with the appropriate example and suggest a solution to it. 10
(CO3)
(PO1)
3. a) State the reasons for introducing foreign keys into the database with example(s). 10
- b) Write SQL DDLs corresponding to schemas given below: 15
- *driver* (*driver id*, *name*, *address*)
 - *vehicle* (*license*, *model*, *year*)
 - *incident* (*report number*, *date*, *location*)
 - *owner* (*driver id*, *license*)
 - *damage* (*report number*, *license*, *driver id*, *damage amount*)

Make any reasonable assumptions about data types and be sure to declare primary and foreign keys.

4. State the normalization goal to ensure good design of a database using an example. 5